



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

**Salem Terminal TO Tiruchengodu Rig Urimayalargal
Sang**

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

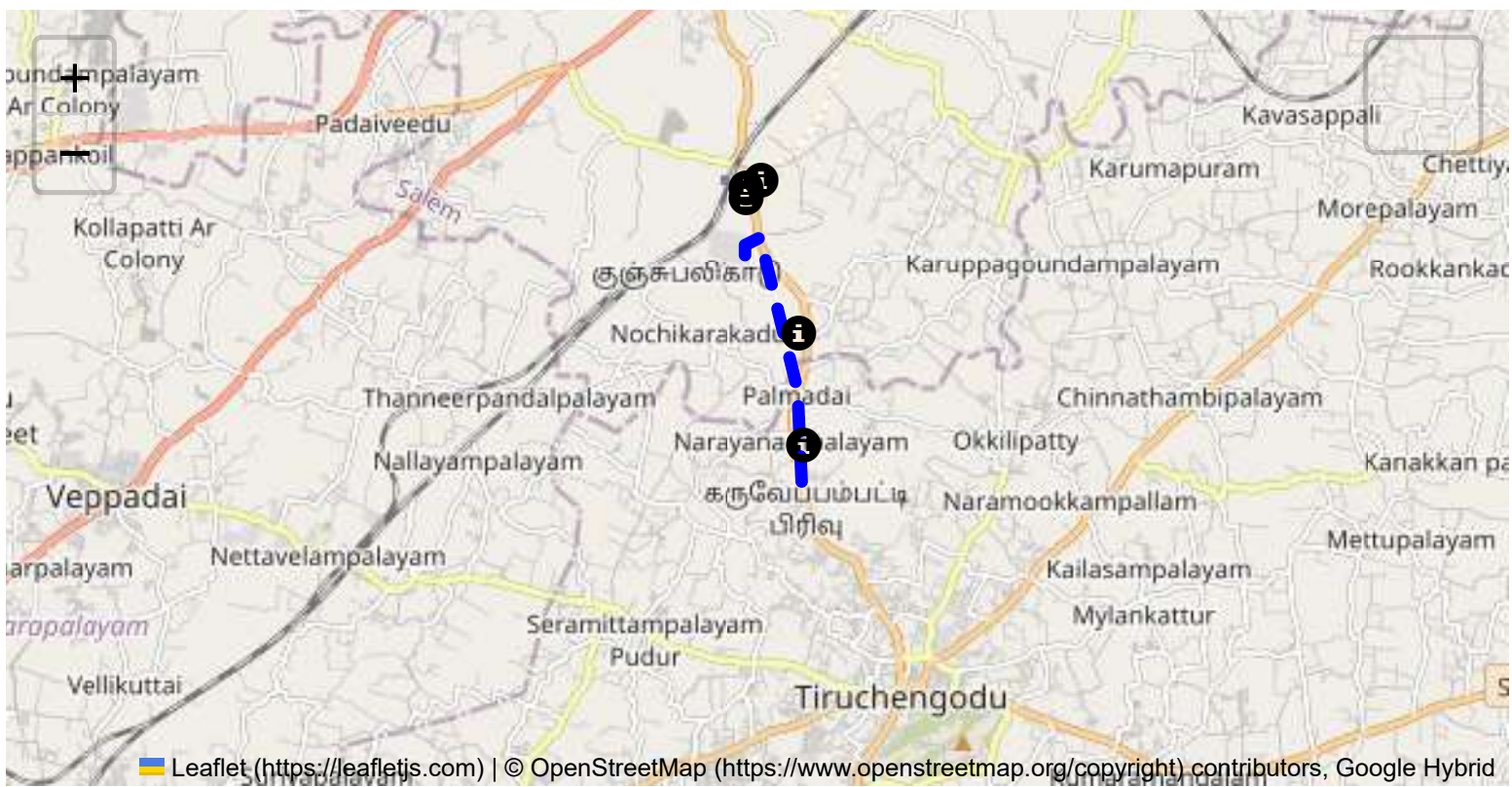
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

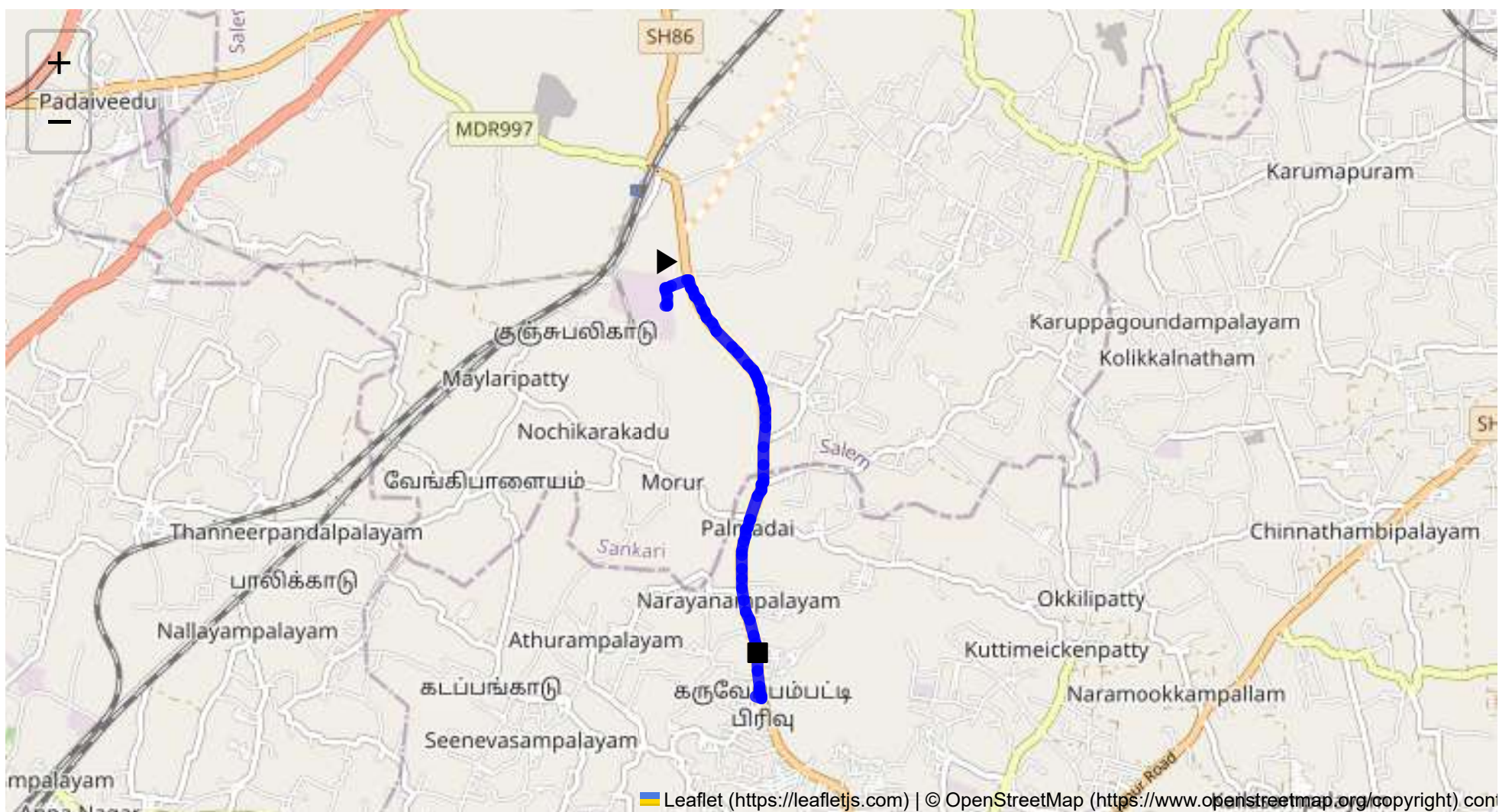
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 4.73 km
Estimated Duration: 0.1 hours
Adjusted Duration (Heavy Vehicle): 0.2 hours
Start: (11.4381, 77.8734)
End: (11.404666, 77.881302)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Karuveppampatti via Puthur, Kangeyampalayam, and Palmadai covers approximately 4.73 kilometers primarily along local roads. Starting from Sangagiri, the route progresses eastwards, connecting SH 86 and other minor roads. This is a rural area with a mix of small townships and agricultural land.

2. Typical Weather Conditions and Potential Hazards

Tamil Nadu generally experiences a tropical climate. Expect hot and humid conditions, with the possibility of heavy rainfall during the monsoon season (October - December). These conditions can lead to slippery roads and reduced visibility, particularly in rural areas. During the summer months (March - June), high temperatures can impact vehicle performance.

3. Analysis of Traffic Patterns

Traffic is generally light to moderate in this region, with increased activity during the early morning (7-9 AM) and late afternoon (5-7 PM), when local residents commute for work or school. Roadside vendors and agricultural vehicles can occasionally cause minor delays. The SH 86 segment may experience higher traffic density due to its use as a regional connector.

4. Assessment of Road Quality and Infrastructure

The roads vary in quality, with state highways like SH 86 typically better maintained than rural roads. Puthur and Kangeyampalayam may have narrow roads with limited shoulder areas, posing challenges for large vehicles. Occasional potholes and uneven surfaces are common, especially after rainfall.

5. Suggestions for Alternative Routes

In case of emergencies, an alternative route via the main arterial roads connecting nearby towns like Edappadi or Kumarapalayam can be considered, although these may significantly extend travel distance and time.

6. Summary of Local Regulations Affecting Hazardous Material Transport

The transport of hazardous materials in Tamil Nadu is governed by the Motor Vehicles Act and the Central Motor Vehicles Rules. There may be restrictions on transporting such materials during certain times, particularly through populated areas. Drivers must ensure all documentation is current and that vehicles are properly marked.

7. Overview of Historical Incidents

There have been limited reports of major incidents involving heavy vehicles on this specific short route. However, general risks include rollover accidents and spillages in areas with sharp curves or sudden inclines.

8. Environmental Considerations and Sensitive Areas

The route traverses predominantly agricultural land; care should be taken to prevent contamination of water bodies and farmland. There are no known environmentally sensitive areas such as wildlife sanctuaries directly on this route.

9. Analysis of Communication Coverage

Mobile communication coverage is generally good along this route with periodic dead zones in the more rural stretches, particularly around Kangeyampalayam. Drivers should be prepared for potential connectivity issues.

10. Estimated Emergency Response Times

Emergency services such as local fire and medical facilities are available in Sangagiri and other nearby towns. Response times can range from 15 to 30 minutes, depending on location and traffic conditions.

12. Overall Summary of Risk Assessment

- Road Quality:** Moderate, with potential hazards due to narrow roads and occasional poor surface conditions.
- Weather Conditions:** Tropical, with potential for hazardous driving conditions during monsoon.
- Traffic:** Low to moderate, with peak hours in the morning and evening.
- Regulations and Safety:** Must comply with hazardous material transport laws; ensure proper documentation and vehicle marking.
- Emergency Services:** Reasonably accessible but with variable response times.

Overall, the route poses minimal to moderate risk, primarily related to road infrastructure quality and weather conditions. Regular checks and adherence to safety guidelines are recommended to ensure safe transport of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43961, 77.87341	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Blind Spot	Blind Spot	11.40452, 77.88181	10 KM/Hr

Route Photos of Risky Spots



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.40452, 77.88181
